Integration by Substitution

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: u-substitution**

Evaluate each integral using substitution:

a $\int 2x(x^2+1)^5\,dx$ $u = x^2+1$: $= \frac{(x^2+1)^6}{6} + C.$

b $\int \frac{x}{\sqrt{x^2+9}}\,dx$ $u = x^2+9$: $= \sqrt{x^2+9} + C.$

c $\int \cos(3x)\,dx$ $u = 3x$: $= \frac{1}{3}\sin(3x) + C.$

(a) $u = x^2+1$: $= \frac{(x^2+1)^6}{6} + C.$; (b) $u = x^2+9$: $= \sqrt{x^2+9} + C.$; (c) $u = 3x$: $= \frac{1}{3}\sin(3x) + C.$

2 **[Foundational]** • **Target: Definite integrals with substitution**

Evaluate $\int_0^2 x\sqrt{x^2+1}\,dx$ exactly, changing the limits with $u = x^2+1$.

u goes from 1 to 5: $\frac{1}{2}\int_1^5 \sqrt{u}\,du = \frac{1}{2}\left[\frac{2}{3}u^{3/2}\right]_1^5 = \frac{1}{3}(5\sqrt{5} - 1).$

3 **[Foundational]** • **Target: Substitution with trigonometric and exponential integrands**

Evaluate $\int \frac{2x+3}{x^2+3x+1}\,dx$, recognizing the numerator as the derivative of the denominator.

$u = x^2+3x+1$, $du = (2x+3)\,dx$: $= \ln|x^2+3x+1| + C.$

4 **[Foundational]** • **Target: Choosing the substitution**

For each integral, state a suitable substitution u and evaluate:

a $\int x^2(x^3-1)^7\,dx$ $u = x^3-1$, $du = 3x^2dx$: $= \frac{1}{3}\int u^7\,du = \frac{(x^3-1)^8}{24} + C.$

b $\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx$ $u = \cos x$, $du = -\sin x\,dx$: $= -\ln|\cos x| + C = \ln|\sec x| + C.$

(a) $u = x^3-1$, $du = 3x^2dx$: $= \frac{1}{3}\int u^7\,du = \frac{(x^3-1)^8}{24} + C.$; (b) $u = \cos x$, $du = -\sin x\,dx$: $= -\ln|\cos x| + C = \ln|\sec x| + C.$

5 **[Foundational]** • **Target: Substitution with radicals**

Evaluate $\int x\sqrt{2-x}\,dx$ using the substitution $u = 2-x$ (so $x = 2-u$).

$u = 2-x$, $x = 2-u$, $dx = -du$: $\int (2-u)\sqrt{u}(-du) = \int (u-2)\sqrt{u}\,du = \int (u^{3/2} - 2u^{1/2})du = \frac{2}{5}u^{5/2} - \frac{4}{3}u^{3/2} + C = \frac{2}{5}(2-x)^{5/2} - \frac{4}{3}(2-x)^{3/2} + C.$

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6 [Foundational] • Target: Substitution with radicals

Evaluate $\int_0^3 \frac{x}{\sqrt{x+1}}\,dx$ using $u = x+1$ (so $x = u-1$), changing the limits.

u goes from 1 to 4: $\int_1^4 \frac{u-1}{\sqrt{u}}du = \int_1^4 (u^{1/2} - u^{-1/2})du = \left[\frac{2}{3}u^{3/2} - 2u^{1/2}\right]_1^4 = \left(\frac{16}{3} - 4\right) - \left(\frac{2}{3} - 2\right) = \frac{4}{3} + \frac{4}{3} = \frac{8}{3}$.

7 [Foundational] • Target: Visualizing an area found by substitution

The graph shows $f(x) = x\sqrt{x^2+1}$ on $[0, 2]$. Evaluate $\int_0^2 x\sqrt{x^2+1}\,dx$ exactly using $u = x^2+1$, and match it to the shaded area.

u goes from 1 to 5: $\frac{1}{2}\int_1^5 \sqrt{u}\,du = \frac{1}{2}\left[\frac{2}{3}u^{3/2}\right]_1^5 = \frac{1}{3}\left(5\sqrt{5} - 1\right) \approx 3.393$, which equals the shaded area under the curve.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Substitution with double angle and reciprocal forms

Evaluate $\int \sin x \cos x\,dx$ two ways: (a) using $u = \sin x$, and (b) using the identity $\sin x \cos x = \frac{1}{2}\sin(2x)$. Confirm both give equivalent answers.

(a) $u = \sin x$: $\int u\,du = \frac{\sin^2 x}{2} + C$. (b) $\int \frac{1}{2}\sin(2x)\,dx = -\frac{1}{4}\cos(2x) + C$. Using $\cos(2x) = 1 - 2\sin^2 x$: $-\frac{1}{4}(1 - 2\sin^2 x) + C = \frac{\sin^2 x}{2} + C'$ — the same family, differing only by the constant.

9 [Intermediate] • Target: Substitution with double angle and reciprocal forms

Evaluate $\int \frac{1}{x \ln x}\,dx$ using $u = \ln x$.

$u = \ln x$, $du = \frac{1}{x}dx$: $\int \frac{1}{u}\,du = \ln|u| + C = \ln|\ln x| + C$.

10 [Intermediate] • Target: Substitution in motion and rate problems

A particle's velocity is $v(t) = t\sin(t^2)$ m/s. Find its displacement from $t = 0$ to $t = \sqrt{\pi}$, using $u = t^2$.

$u = t^2$, $du = 2t\,dt$, limits 0 to π: $\frac{1}{2}\int_0^\pi \sin u\,du = \frac{1}{2}[-\cos u]_0^\pi = \frac{1}{2}(1+1) = 1$ m.

11 [Intermediate] • Target: Recognizing when substitution simplifies a problem

Evaluate $\int (x+1)\sqrt{x^2+2x+5}\,dx$ by noticing the numerator is (up to a constant) the derivative of the expression under the root.

$u = x^2+2x+5$, $du = (2x+2)dx = 2(x+1)dx$: $\frac{1}{2}\int \sqrt{u}\,du = \frac{1}{2}\cdot\frac{2}{3}u^{3/2} + C = \frac{1}{3}(x^2+2x+5)^{3/2} + C$.

12 [Intermediate] • Target: Substitution with trigonometric and exponential integrands

[Techniques of Integration · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

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13 **[Intermediate]** • **Target: Choosing the substitution**

[Partial Derivatives & the Gradient · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

14 **[Intermediate]** • **Target: Substitution with radicals**

Use the substitution $x = 2\sin\theta$ to evaluate the exact definite integral $\int_0^1 \sqrt{4 - x^2}\, dx$.

1. Substitute $x = 2\sin\theta \Rightarrow dx = 2\cos\theta\, d\theta$. When $x = 0$, $\theta = 0$. When $x = 1$, $2\sin\theta = 1 \Rightarrow \sin\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$. 2. Rewrite integrand: $\sqrt{4 - x^2} = \sqrt{4 - 4\sin^2\theta} = 2\sqrt{\cos^2\theta} = 2\cos\theta$ (since $\cos\theta > 0$ on $[0, \pi/6]$). 3. Substitute into the integral.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Visualizing an area found by substitution** • **Cross-Lesson: 2.01 Evaluating Limits**

Level 1 · FoundationsFind the area of the region bounded by $x = y^2$ and $x = y + 2$ using integration with respect to y.

Find intersections in terms of y: $y^2 = y + 2 \Rightarrow y^2 - y - 2 = 0 \Rightarrow (y - 2)(y + 1) = 0 \Rightarrow y = -1,\ y = 2$ For $y \in (-1, 2)$, test $y = 0$: $x_{\text{line}} = 2 > x_{\text{parabola}} = 0$. The right curve is $x = y + 2$ and the left curve is $x = y^2$. Integrate with respect to y: $A = \int_{-1}^{2} [(y + 2) - y^2]\, dy = \left[\frac{y^2}{2} + 2y - \frac{y^3}{3}\right]_{-1}^{2}$ At $y = 2$: $2 + 4 - \frac{8}{3} = 6 - \frac{8}{3} = \frac{10}{3}$. At $y = -1$.

16 **[Challenging]** • **Target: Substitution with double angle and reciprocal forms** • **Cross-Lesson: 2.02 Continuity**

Level 2 · MediumA 13-foot ladder leans against a vertical wall. The base of the ladder slides away from the wall at a constant rate of 2 ft/s. How fast is the top of the ladder sliding down the wall when the base is 5 feet from the wall?

Let x be the distance from the base to the wall and y be the height on the wall. By the Pythagorean theorem: $x^2 + y^2 = 13^2 = 169$ At the instant $x = 5$: $y = \sqrt{169 - 25} = \sqrt{144} = 12$ ft. Differentiating implicitly with respect to time t: $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow x\frac{dx}{dt} + y\frac{dy}{dt} = 0$ Substitute $x = 5$, $y = 12$, and $\frac{dx}{dt} = 2$ ft/s.

17 **[Challenging]** • **Target: Substitution in motion and rate problems** • **Cross-Lesson: 3.01 The Derivative from First Principles**

Use the substitution $u = 2x + 1$ to find the exact value of $\int_0^4 \frac{x}{\sqrt{2x + 1}}\, dx$.

Let $u = 2x + 1$, so $du = 2\, dx \Rightarrow dx = \frac{1}{2}\, du$, and $x = \frac{u - 1}{2}$. Change limits: $x = 0 \Rightarrow u = 1, \quad x = 4 \Rightarrow u = 9$ Transform the integral: $\int_0^4 \frac{x}{\sqrt{2x + 1}}\, dx = \int_1^9 \frac{\frac{u-1}{2}}{\sqrt{u}} \cdot \frac{1}{2}\, du = \frac{1}{4}\int_1^9 (u^{1/2} - u^{-1/2})\, du$ $= \frac{1}{4}\left[\frac{2}{3}u^{3/2} - 2u^{1/2}\right]_1^9 = \frac{1}{4}\left(\left(\frac{2}{3}(27) - 2(3)\right) - \left(\frac{2}{3}(1) - 2(1)\right)\right)$.

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18 [Challenging] • Target: Recognizing when substitution simplifies a problem • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Use trigonometric substitution to evaluate the indefinite integral $\int \frac{1}{(4+x^2)^{3/2}}\,dx$.

1. Substitute $x = 2\tan\theta \Rightarrow dx = 2\sec^2\theta\,d\theta$. Then $4 + x^2 = 4 + 4\tan^2\theta = 4\sec^2\theta$, so: $(4+x^2)^{3/2} = (4\sec^2\theta)^{3/2} = 8\sec^3\theta$. 2. Substitute into the integral: $\int \frac{2\sec^2\theta}{8\sec^3\theta}\,d\theta = \frac{1}{4}\int \frac{1}{\sec\theta}\,d\theta = \frac{1}{4}\int \cos\theta\,d\theta = \frac{1}{4}\sin\theta + C$. 3. Express $\sin\theta$ in terms of x: From $\tan\theta = \frac{x}{2}$, construct a right triangle with opposite side x, adjacen.

19 [Challenging] • Target: u-substitution • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Level 3 · AdvancedLet R be the region bounded by $y = \sqrt{x}$, $y = 2$, and the y-axis. Set up, but do not evaluate, an integral for the volume generated when R is revolved about the vertical line $x = -1$.

1. The region has boundaries $x = y^2$ (right boundary), $x = 0$ (left boundary / y-axis), and y runs from $y = 0$ to $y = 2$. 2. The axis of rotation is the vertical line $x = -1$. Slicing perpendicular to the axis of rotation requires integration with respect to y (horizontal washers). 3. Determine outer and inner radii:Outer radius (distance from $x = -1$ to furthest curve $x = y^2$): $R(y) = y^2 - (-1) = y^2 + 1$ Inner radius (distance from $x = -1$ to closer curve $x = 0$): $r(y) = 0 - (-1) = 1$ 4. Set up.

20 [Challenging] • Target: Definite integrals with substitution • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Level 2 · MediumFind the angle(s) $\theta \in [0, \pi]$ at which the polar curve $r = 3\cos\theta$ passes through the origin, and determine the equation of the tangent line at that point.

Set $r = 0$: $3\cos\theta = 0 \Rightarrow \theta = \frac{\pi}{2}$. By the Tangents at the Pole Theorem, when $r(\theta_0) = 0$ and $r'(\theta_0) \neq 0$, the line $\theta = \theta_0$ is the tangent line. Thus, the tangent line at the pole is $\boldsymbol{\theta = \frac{\pi}{2}}$ (the vertical line $x = 0$).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: u-substitution

A ladder 13 m long leans against a wall. Base slides at 2 m/s. Rate of top falling when base is 5 m from wall:

a. $-\frac{5}{6}$ m/s ✓

b. $-\frac{5}{12}$ m/s

c. $-\frac{2}{5}$ m/s

d. $-\frac{1}{2}$ m/s

From $x^2 + y^2 = 169$, when $x = 5$ we have $y = 12$. Differentiating gives $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{5(2)}{12} = -\frac{5}{6}$ m/s.

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22 [Foundational] • Target: Definite integrals with substitution

Area between $y = x$ and $y = x^2$: intersect at $x =$

a. 0 **and** 1 ✓

b. 1 and 2

c. 0 and 2

d. -1 and 1

Applying Applications of Integration principles, $x = x^2 \Rightarrow x(x-1) = 0$. Therefore, the verified correct option is 0 and 1.

23 [Foundational] • Target: Substitution with trigonometric and exponential integrands

Find the total area enclosed by the cardioid $r = 2(1 - \cos\theta)$.

a. 6π ✓

b. 3π

c. 4π

d. 8π

The polar area is $A = \frac{1}{2}\int_0^{2\pi} 4(1 - 2\cos\theta + \cos^2\theta)\,d\theta = 2(2\pi + \pi) = 6\pi$.

24 [Foundational] • Target: Choosing the substitution

Fluid force: pressure at depth d is $\rho g d$. Force on area A:

a. $\rho g A$

b. ρA^2

c. $\rho g d A$ ✓

d. $g d$

Applying Applications of Integration principles, $F = PA = \rho g d A$. Therefore, the verified correct option is $\rho g d A$.

25 [Foundational] • Target: Substitution with radicals

Surface area of $y = \sqrt{x}$ revolved about the x-axis from 0 to 4:

a. $\frac{\pi}{3}(17\sqrt{17} - 1)$

b. $\pi(\sqrt{17} - 1)$

c. 8π

d. $\frac{\pi}{6}(17\sqrt{17} - 1)$ ✓

$S = 2\pi\int_0^4 \sqrt{x}\sqrt{1 + 1/(4x)}\,dx = \pi\int_0^4 \sqrt{4x+1}\,d(x) \to \frac{\pi}{6}(17\sqrt{17} - 1)$.

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26 [Foundational] • Target: Visualizing an area found by substitution

Volume of solid of revolution when region $0 \le y \le e^{-x^2}$, $0 \le x \le 1$ is rotated about y-axis (shell):

a. $\pi(1 - e^{-1})$

b. $\pi(e - 1)/e$

c. $\pi/2$

d. $2\pi\int_0^1 xe^{-x^2}dx$ ✓

Shell: $V = 2\pi\int_0^1 xe^{-x^2}dx = \pi(1 - e^{-1})$. Choice C is the setup (correct), choice A = the value.

27 [Foundational] • Target: Substitution with double angle and reciprocal forms

Surface area of Gabriel's horn $y = 1/x$, $x \ge 1$, rotated about x-axis:

a. **diverges** ✓

b. π

c. 2π

d. finite

$S = 2\pi\int_1^\infty (1/x)\sqrt{1 + 1/x^4}\,dx > 2\pi\int_1^\infty (1/x)dx$: diverges (Painter's paradox).

28 [Foundational] • Target: Substitution in motion and rate problems

Area enclosed by $r = \sin 2\theta$ (four-petal rose):

a. $\pi/4$

b. $\pi/2$ ✓

c. π

d. 2π

Each petal area $= \frac{1}{2}\int_0^{\pi/2} \sin^2(2\theta)d\theta = \pi/8$; four petals: $\pi/2$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Recognizing when substitution simplifies a problem

Exchange integration order: $\int_0^4\int_{\sqrt{x}}^2 f(x,y)\,dy\,dx =$

a. $\int_0^4\int_0^y f\,dx\,dy$

b. $\int_0^2\int_0^y f\,dx\,dy$

c. $\int_0^2\int_y^4 f\,dx\,dy$

d. $\int_0^2\int_0^{y^2} f(x,y)\,dx\,dy$ ✓

Region: $0 \le x \le 4, \sqrt{x} \le y \le 2 \Leftrightarrow 0 \le y \le 2, 0 \le x \le y^2$.

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30 [Intermediate] • Target: u-substitution

Region bounded by $y = 3 - x^2$ and $y = -1$. Area:

a. 8

b. 16/3

c. 4

d. 32/3 ✓

Intersect at $x = \pm 2$. $\int_{-2}^{2}(4 - x^2)dx = 2\int_{0}^{2} = 2[4x - x^3/3]_0^2 = 2(8 - 8/3) = 32/3$.

31 [Intermediate] • Target: Definite integrals with substitution

Find the area of the surface formed by revolving $y = 4x$, $0 \le x \le 2$, about the x-axis.

a. $8\sqrt{17}\,\pi$

b. 16π

c. $\frac{128\pi}{3}$

d. $16\sqrt{17}\,\pi$ ✓

$S = 2\pi\int_0^2 4\sqrt{17}\,x\,dx \;=\; 16\sqrt{17}\,\pi.$

32 [Intermediate] • Target: Substitution with trigonometric and exponential integrands

Area of the region enclosed by the lemniscate $r^2 = 4\cos 2\theta$:

a. 4 ✓

b. 2

c. 8

d. π

$A = \frac{1}{2}\int_{-\pi/4}^{\pi/4} 4\cos 2\theta \cdot 2\,d\theta = 4.$

33 [Intermediate] • Target: Choosing the substitution

The equation $y' = ky$ models exponential growth when:

a. $k = 0$

b. $k < 0$

c. $k = 1$

d. $k > 0$ ✓

Applying Differential Equations principles, $k > 0$: growth; $k0$.

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34 [Intermediate] • Target: Substitution with radicals

Integration by parts: $\int xe^x\,dx =$

a. $e^x(x-1)+C$ ✓

b. $e^x(x+1)+C$

c. xe^x+C

d. $\frac{x^2e^x}{2}+C$

IBP: $u=x, dv=e^xdx$: $xe^x-\int e^xdx = xe^x-e^x+C = e^x(x-1)+C$.

35 [Intermediate] • Target: Visualizing an area found by substitution

Area of surface $z=\sqrt{4-x^2-y^2}$ over disk $x^2+y^2\le 4$:

a. 4π

b. π

c. 8π ✓

d. 2π

Upper hemisphere radius 2: surface area $=2\pi(2)^2=8\pi$.

36 [Intermediate] • Target: Substitution with double angle and reciprocal forms

Area between $y=x^2$ and $y=x$ equals:

a. 1/3

b. 1/6 ✓

c. 1/2

d. 1

Intersect at 0,1. $\int_0^1(x-x^2)dx = 1/2-1/3 = 1/6$.

37 [Intermediate] • Target: Substitution in motion and rate problems

MVT for double integrals: $\iint_D f\,dA = f(c,d)\cdot A(D)$ requires f to be:

a. **continuous on a connected region** ✓

b. positive

c. analytic

d. harmonic

MVT for integrals: continuous on connected region.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Recognizing when substitution simplifies a problem • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Improper double integral: $\int_0^\infty \int_0^\infty e^{-(x+y)} dx\, dy =$

a. 1 ✓

b. 0

c. ∞

d. e

$= \left(\int_0^\infty e^{-x} dx\right)^2 = 1^2 = 1.$

39 [Challenging] • Target: u-substitution • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

For $f \geq 0$, the double integral represents:

a. volume below $z = 0$

b. **volume between the surface $z = f(x, y)$ and the xy-plane** ✓

c. area of the surface

d. perimeter

Non-negative integrand: volume interpretation.

40 [Challenging] • Target: Definite integrals with substitution • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Fubini's theorem allows switching the order of integration for:

a. all functions

b. only constant functions

c. **non-negative or absolutely integrable functions** ✓

d. periodic functions

Fubini: valid for non-negative or $f \in L^1$.

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41 [Challenging] • Target: Substitution with trigonometric and exponential integrands • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Green's theorem area formula for ellipse $x = a\cos t, y = b\sin t$:

a. ab

b. $2\pi ab$

c. $\pi(a+b)$

d. πab ✓

Area $= \frac{ab}{2} \cdot 2\pi = \pi ab$.

42 [Challenging] • Target: Choosing the substitution • Cross-Lesson: 6.01 Related Rates

The region between $y = e^x$ and $y = 1$ from $x = 0$ to $x = 1$ has area:

a. 1

b. $e - 1$

c. e

d. $e - 2$ ✓

$\int_0^1 (e^x - 1)dx = [e^x - x]_0^1 = (e-1) - 1 = e - 2$.

43 [Challenging] • Target: Substitution with radicals • Cross-Lesson: 6.02 Optimization and Curve Sketching

Polar tangent: the curve passes through origin when $r = 0$. Tangent angle $\psi =$

a. π

b. θ_0 **where** $r(\theta_0) = 0$ ✓

c. 0

d. $\pi/2$

At origin, tangent direction $= \theta_0$.

44 [Challenging] • Target: Visualizing an area found by substitution • Cross-Lesson: 7.02 Integration by Parts

Separable double integral: if $f(x,y) = g(x)h(y)$, then $\int_a^b \int_c^d f\,dy\,dx =$

a. $\int_a^b g \int_c^d h$

b. $\int_a^b gh\,dx$

c. $\int_a^b g + \int_c^d h$

d. $(\int_a^b g)(\int_c^d h)$ ✓

Separable: product of separate integrals.

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45 [Challenging] • Target: Substitution with double angle and reciprocal forms • Cross-Lesson: 8.01 Area Between Curves

Area between $y = f(x)$ and x-axis on $[a, b]$ equals:

a. $\int_a^b |f(x)|\,dx$ ✓

b. $\int_a^b f(x)\,dx$ only when $f \geq 0$

c. $f(b) - f(a)$

d. $\int_a^b f(x)^2\,dx$

Always use absolute value for area.

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